



1ST INTERNATIONAL CONFERENCE ON INTELLIGENT SIGNAL PROCESSING AND EFFECTIVE COMMUNICATION TECHNOLOGIES (INSPECT-2024)

“Technological Innovations for a Better World !”

ABV-IIITM Gwalior, India, 07-08 December 2024

ABOUT INSPECT-2024 (HYBRID MODE)

Welcome to the 2024 International Conference on Intelligent Signal Processing and Effective Communication Technologies (INSPECT-2024), to be held in December 2024 at ABV-IIITM Gwalior, India. The conference is the premier forum for presenting new advances and research results in varied disciplines and their associated applications in all aspects of Communication Technology, Signal Processing Informatics, and Machine Learning that consider improving and developing quality of human life and better living on earth. In addition to the technical sessions, there will be invited sessions, panel sessions, and keynote addresses. The aim of the conference is to stimulate interaction and convergent among researchers active in the areas of Communication Technology, Signal Processing Informatics, and Machine Learning. The conference will be sponsored by the IEEE MP Section.

ABOUT ABV-IIITM GWALIOR

Established in 1997, with a financial outlay of Rs. 61.0 Crores (the US \$14 million) for the first five years, the institute serves as a premier institution in the field of IT and management. Over the years the institute has been providing highly respected Techno-Managers to the IT industry doing various techno-functional roles with elan. The institute was accorded the status of deemed university in the year 2001. The institute campus is spread over 160 acres (0.65 sq. km) amidst a picturesque landscape in the foothills of the Gwalior Fort. The campus is self-contained and equipped with all the state-of-the-art facilities necessary for the all-round development of the student's personalities.

The Department of Electrical and Electronics Engineering is one of the major departments being created in 2022 to achieve the mission and vision of the institute. The Department is currently in its development phase and engaged in teaching, research and development in the areas of Wireless communication, multimedia signal processing, machine learning, power systems, VLSI, IoT devices, autonomous control systems, high-frequency devices, and communication systems. Department of Electrical and Electronics Engineering is organizing INSPECT-2024.

CONFERENCE TRACKS

INSPECT-2024 welcomes high-quality, original, and previously unpublished submissions in computer vision and machine intelligence. Topics of interests for the INSPECT-2024 conference include, but not limited to, the following:

Track-1: Signal & Image Processing

- Array Signal Processing
- Nonlinear Signal Processing
- Adaptive Signal Processing
- Audio Signal Processing
- Speech Processing
- Digital Signal Processing
- Image Enhancement and Restoration
- Image Compression
- Pattern Recognition
- Computer Vision
- Medical Image Processing
- Biomedical Signal Processing
- Remote Sensing
- Multimedia Signal Processing
- Sparse Signal Processing
- Hyperspectral Imaging
- Geospatial Image Processing
- Biometric Signal and Image Processing
- Deep Learning for Signal Processing
- Real-Time Signal and Image Processing
- Acoustic Sensor Network

Track-2: Communication Systems & Networking

- Wireless Communication Systems
- 5G and Beyond
- Satellite Communication Systems
- Network Security
- Cognitive Radio Networks
- Software-Defined Networking (SDN)
- Information Theory
- Cloud Computing
- Network Protocols
- Communication Hardware Design
- Social Networks Analysis
- Multimedia Communication Systems
- Network Virtualization
- Quantum Communication
- Network Performance Optimization
- Green Networking Technologies
- Deep Learning for communication
- Edge Computing

Track-3: IoT, Smart Grid, Robotics & Control Systems

- Internet of Things (IoT)
- Smart Grid Technologies
- Industrial Automation
- Robotics and Automation
- Control Systems Engineering
- Cyber-Physical Systems
- Energy Management Systems
- Sensor Networks
- Autonomous Vehicles
- Human-Robot Interaction
- Distributed Control Systems
- Embedded Systems Design
- Renewable Energy Integration
- Smart Cities Infrastructure
- Home Automation
- Intelligent Transportation Systems
- Swarm Robotics, Wearable Technology
- Intelligent Robotics and Control

Track-4: RF, Microwave & Optical Communication

- Smart Radio Frequency Engineering
- Microwave Engineering
- Antenna Design
- Radar Systems
- Millimeter-Wave Communication
- Optical Fiber Communication
- Free-Space Optical Communication
- Microwave Photonics
- RF MEMS
- Optical Switching and Routing
- RF Circuit Design
- Optical Networks
- Microwave Remote Sensing
- Terahertz Communication
- Photonic Integrated Circuits

Track-5: VLSI for Signal Processing & Communication

- Digital VLSI Design
- Analog VLSI Design
- Mixed-Signal VLSI Design
- Low-Power VLSI Design
- ASIC Design
- Neuromorphic Engineering
- VLSI CAD (Computer-Aided Design)
- High-Level Synthesis
- Hardware Acceleration
- VLSI Design for Signal Processing
- Hardware Security
- Reconfigurable Computing
- FPGA Design
- VLSI Testing and Verification
- SoC (System-on-Chip) Design
- Advanced CMOS Technologies
- Clock and Power Distribution Network
- VLSI Architectures for Communication

IMPORTANT DATES

Call for Paper	15 April 2024	Publication: The proceedings of the conference will be published by IEEE (Scopus indexed). Accepted papers will be submitted for inclusion into IEEE Xplore subject to meeting IEEE Xplore's scope and quality requirements. Any paper selected for presentation but not presented at the conference will not be published by IEEE.
Submission Deadline	31 July 2024	
Notification of Acceptance	10 September 2024	
Camera Ready Submission Deadline	25 September 2024	
Early Bird Registration	10 October 2024	
Registration Close	30 October 2024	
INSPECT-2024	07-08 December 2024	

SUBMISSION GUIDELINES

Submission Guidelines Prospective authors are invited to submit full-length original research papers. In submitting a manuscript to INSPECT-2024, the authors acknowledge that no paper substantially similar in content has been or will be submitted to another journal, conference, or workshop during the review period. In such a case, the paper will be rejected without review. Papers must be electronically submitted without exception before the deadline expires. The paper must be at most 6 pages in length (including all text, figures, and references). However, authors can add up to 2 extra pages with the appropriate fee payment per page. The manuscript must be submitted in pdf format only, and the file size of your manuscript should not exceed 10 MB. INSPECT-2024 follows a single-blind review process. Authors must use the manuscript template as follows: Overleaf, LaTeX, or Word. The manuscript has to be uploaded online at the INSPECT-2024 using the Microsoft CMT paper submission portal.

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CONTACT US: ☎ +91-9109106995, +91-9586288218

✉ inspect2024@iiit.ac.in 🌐 www.inspect.iiit.ac.in